

# **API Reference**

# SDN Agent to KMS API

**API Version: 2.0.0**

Messages the SDN sends to the KMS

## **CONTACT**

**URL:** <https://github.com/ait-crypto/kms-sdn-agent-interface-specification/issues>

# INDEX

|                                            |           |
|--------------------------------------------|-----------|
| <b>1. API</b>                              | <b>4</b>  |
| 1.1 GET /version                           | 4         |
| <b>2. CONTROL</b>                          | <b>5</b>  |
| 2.1 PUT /control/status                    | 5         |
| 2.2 PUT /control/qkd-device/{device_id}    | 6         |
| 2.3 POST /control/qkd-device/{device_id}   | 7         |
| 2.4 DELETE /control/qkd-device/{device_id} | 8         |
| <b>3. LINK</b>                             | <b>10</b> |
| 3.1 GET /link/performance/{link_id}        | 10        |
| 3.2 PUT /link/relay/{key_stream_id}        | 11        |
| 3.3 POST /link/relay/{key_stream_id}       | 12        |
| 3.4 DELETE /link/relay/{key_stream_id}     | 13        |
| <b>4. MONITOR</b>                          | <b>15</b> |
| 4.1 GET /monitor/capabilities              | 15        |
| 4.2 GET /monitor/device-info               | 15        |
| 4.3 GET /monitor/status                    | 16        |
| 4.4 GET /monitor/log                       | 17        |
| 4.5 GET /monitor/qkd-device                | 18        |

# Security and Authentication

## SECURITY SCHEMES

| KEY   | TYPE      | DESCRIPTION                                                                    |
|-------|-----------|--------------------------------------------------------------------------------|
| https | mutualTLS | Connections must use TLS 1.3. The server enforces this at the transport layer. |

# API

## 1. API

Generic endpoints related with this API

### 1.1 GET /version

#### Get this API version

State this SDN API version number which is implemented and supported by the KMS

### REQUEST

No request parameters

### RESPONSE

**STATUS CODE - 200:** Provided API version

#### RESPONSE MODEL - application/json

| NAME                        | TYPE   | DESCRIPTION                                                                                                                                                                                                                                      |
|-----------------------------|--------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |        |                                                                                                                                                                                                                                                  |
| version*                    | string | <b>PATTERN:</b> ^(0 [1-9]\d*)\.(0 [1-9]\d*)\.(0 [1-9]\d*)(?:-((?:0 [1-9]\d* \d*[a-zA-Z-][0-9a-zA-Z-]*)?(?:\.(?:0 [1-9]\d* \d*[a-zA-Z-][0-9a-zA-Z-]*)*)?)?(?:\+(0-9a-zA-Z-]+(?:\.(0-9a-zA-Z-]+)*)?)?)?\$<br>Version number in semantic versioning |

**STATUS CODE - 400:** Bad request

#### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 500:** Internal error

#### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

## 2. CONTROL

Generic control endpoints

### 2.1 PUT /control/status

#### Modify state of the KMS

Modify state of the KMS, it can be online, stopped or offline.

#### REQUEST

##### REQUEST BODY - application/json

| NAME   | TYPE | DESCRIPTION                      |
|--------|------|----------------------------------|
| state* | enum | ALLOWED:stopped, offline, online |

#### RESPONSE

**STATUS CODE - 200:** changing state successful

**STATUS CODE - 400:** Bad request

##### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 403:** Forbidden, client does not have the required rights.

##### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 500:** Internal error

##### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                              |
|-----------------------------|---------|------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                          |
| type                        | string  | Webpage with information on this problem |
| title*                      | string  | Brief description of the problem         |
| status*                     | integer | Response code or status code             |

| NAME    | TYPE   | DESCRIPTION                                             |
|---------|--------|---------------------------------------------------------|
| detail* | string | Information on the problem and hints on how to solve it |

## 2.2 PUT /control/qkd-device/{device\_id}

### Update an existing QKD device

Update mutable fields of a known QKD device.

## REQUEST

### PATH PARAMETERS

| NAME       | TYPE | EXAMPLE | DESCRIPTION |
|------------|------|---------|-------------|
| *device_id | uuid |         |             |

### REQUEST BODY - application/json

| NAME            | TYPE   | DESCRIPTION                                                                                    |
|-----------------|--------|------------------------------------------------------------------------------------------------|
| label*          | string | Human readable label, for example vendor name and model number.                                |
| state*          | enum   | <b>ALLOWED:</b> online, stopped, offline                                                       |
| interface*      | enum   | <b>ALLOWED:</b> etsi_004, etsi_014, skip, kmip                                                 |
| protocol*       | string | for ETSI GS QKD 014 and SKIP typically https.                                                  |
| peer_kms*       | string | to which peer is this device connected, does not have to be UUID format, but unique in network |
| server_address* | string |                                                                                                |
| primary_id*     | string | In ETSI GS QKD 014 also called Master SAE ID                                                   |
| secondary_id*   | string | In ETSI GS QKD 014 also called Slave SAE ID                                                    |

## RESPONSE

**STATUS CODE - 200:** QKD device updated successfully

**STATUS CODE - 400:** Invalid input (missing device\_id or malformed update object)

### RESPONSE MODEL - application/json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 404:** Not found. This corresponds to the endpoint not found, as 404s can also be generated by the webserver. Use 400 if the device ID string was unknown to KMS

**STATUS CODE - 500:** Server-side error while updating the device

### RESPONSE MODEL - application/json

| NAME                               | TYPE   | DESCRIPTION                              |
|------------------------------------|--------|------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |        |                                          |
| type                               | string | Webpage with information on this problem |

| NAME    | TYPE    | DESCRIPTION                                             |
|---------|---------|---------------------------------------------------------|
| title*  | string  | Brief description of the problem                        |
| status* | integer | Response code or status code                            |
| detail* | string  | Information on the problem and hints on how to solve it |

## 2.3 POST /control/qkd-device/{device\_id}

### Register a new QKD device

Create/register a new QKD device in the KMS instance.

## REQUEST

### PATH PARAMETERS

| NAME       | TYPE | EXAMPLE | DESCRIPTION |
|------------|------|---------|-------------|
| *device_id | uuid |         |             |

### REQUEST BODY - application/json

| NAME            | TYPE   | DESCRIPTION                                                                                    |
|-----------------|--------|------------------------------------------------------------------------------------------------|
| label*          | string | Human readable label, for example vendor name and model number.                                |
| state*          | enum   | <b>ALLOWED:</b> online, stopped, offline                                                       |
| interface*      | enum   | <b>ALLOWED:</b> etsi_004, etsi_014, skip, kmip                                                 |
| protocol*       | string | for ETSI GS QKD 014 and SKIP typically https.                                                  |
| peer_kms*       | string | to which peer is this device connected, does not have to be UUID format, but unique in network |
| server_address* | string |                                                                                                |
| primary_id*     | string | In ETSI GS QKD 014 also called Master SAE ID                                                   |
| secondary_id*   | string | In ETSI GS QKD 014 also called Slave SAE ID                                                    |

## RESPONSE

**STATUS CODE - 201:** QKD device successfully created

**STATUS CODE - 400:** Invalid input (missing fields or malformed body)

### RESPONSE MODEL - application/json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 409:** Device already exists (conflict on device\_id)

### RESPONSE MODEL - application/json

| NAME                               | TYPE   | DESCRIPTION                              |
|------------------------------------|--------|------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |        |                                          |
| type                               | string | Webpage with information on this problem |

| NAME    | TYPE    | DESCRIPTION                                             |
|---------|---------|---------------------------------------------------------|
| title*  | string  | Brief description of the problem                        |
| status* | integer | Response code or status code                            |
| detail* | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 500:** Server-side error while creating the device

#### RESPONSE MODEL - application/json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

## 2.4 DELETE /control/qkd-device/{device\_id}

### Delete QKD device

Remove connected QKD device, KMS will no longer try to obtain keys from it.

### REQUEST

#### PATH PARAMETERS

| NAME       | TYPE | EXAMPLE | DESCRIPTION |
|------------|------|---------|-------------|
| *device_id | uuid |         |             |

### RESPONSE

**STATUS CODE - 204:** QKD device correctly deleted

**STATUS CODE - 400:** Bad request

#### RESPONSE MODEL - application/problem+json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 404:** Not found, API endpoint does not exist.

**STATUS CODE - 500:** Internal error

#### RESPONSE MODEL - application/problem+json

| NAME                               | TYPE   | DESCRIPTION                              |
|------------------------------------|--------|------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |        |                                          |
| type                               | string | Webpage with information on this problem |
| title*                             | string | Brief description of the problem         |



| NAME    | TYPE    | DESCRIPTION                                             |
|---------|---------|---------------------------------------------------------|
| status* | integer | Response code or status code                            |
| detail* | string  | Information on the problem and hints on how to solve it |

## 3. LINK

Monitoring and control related with the link

### 3.1 GET /link/performance/{link\_id}

#### Get performance data for link

Get the performance data for the link with the given link id. Query parameter for entries is optional.

#### REQUEST

##### PATH PARAMETERS

| NAME     | TYPE | EXAMPLE | DESCRIPTION |
|----------|------|---------|-------------|
| *link_id | uuid |         | link's UUID |

##### QUERY PARAMETERS

| NAME    | TYPE    | EXAMPLE | DESCRIPTION                                                                                                                                     |
|---------|---------|---------|-------------------------------------------------------------------------------------------------------------------------------------------------|
| entries | integer |         | Optional. Number of history records requested. If not specified maximal available history is returned. If 0, only current value is transmitted. |

#### RESPONSE

**STATUS CODE - 200:** The response contains KMS performance data

##### RESPONSE MODEL - application/json

| NAME                               | TYPE    | DESCRIPTION                                                                                                                        |
|------------------------------------|---------|------------------------------------------------------------------------------------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                                                                                                    |
| <b>current*</b>                    | object  |                                                                                                                                    |
| timestamp*                         | string  | ISO 8601 format                                                                                                                    |
| delta_t*                           | integer | <b>DEFAULT:10</b><br>Time period over which keys are counted for the rate calculation. The unit is milliseconds.                   |
| IKR*                               | integer | Internal Key Rate = amount of key material reserved for internal use during a delta_t period divided by delta_t. Given in Byte/sec |
| EXKR*                              | integer | External Key Rate = amount of key material reserved for external use during a delta_t period divided by delta_t. Given in Byte/sec |
| SKR*                               | integer | Secret Key Rate = amount of key material received from QKD devices during a delta_t period divided by delta_t. Given in Byte/sec   |
| KAV*                               | integer | Key Availability = amount of key material in Byte available to Applications during a delta_t period                                |
| <b>history</b>                     | array   |                                                                                                                                    |
| timestamp*                         | string  | ISO 8601 format                                                                                                                    |

| NAME     | TYPE    | DESCRIPTION                                                                                                                        |
|----------|---------|------------------------------------------------------------------------------------------------------------------------------------|
| delta_t* | integer | <b>DEFAULT:10</b><br>Time period over which keys are counted for the rate calculation. The unit is milliseconds.                   |
| IKR*     | integer | Internal Key Rate = amount of key material reserved for internal use during a delta_t period divided by delta_t. Given in Byte/sec |
| EXKR*    | integer | External Key Rate = amount of key material reserved for external use during a delta_t period divided by delta_t. Given in Byte/sec |
| SKR*     | integer | Secret Key Rate = amount of key material received from QKD devices during a delta_t period divided by delta_t. Given in Byte/sec   |
| KAV*     | integer | Key Availability = amount of key material in Byte available to Applications during a delta_t period                                |

**STATUS CODE - 400:** Bad request

#### RESPONSE MODEL - application/problem+json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 404:** Not found, API endpoint does not exist

**STATUS CODE - 500:** Internal error

#### RESPONSE MODEL - application/problem+json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

## 3.2 PUT /link/relay/{key\_stream\_id}

### Update a link relay configuration

Replace a link relay configuration with the data in the request, using the given key\_stream\_id as an identifier. Only the data to be changed should be included in data body, supporting partial updates.

## REQUEST

#### PATH PARAMETERS

| NAME           | TYPE | EXAMPLE | DESCRIPTION          |
|----------------|------|---------|----------------------|
| *key_stream_id | uuid |         | key_stream_id's UUID |

#### REQUEST BODY - application/json

| NAME | TYPE | DESCRIPTION |
|------|------|-------------|
|------|------|-------------|

| NAME           | TYPE   | DESCRIPTION              |
|----------------|--------|--------------------------|
| source         | string | Source APP's URI or UUID |
| destination    | array  |                          |
| next_node_info | array  |                          |
| qos_request    | object |                          |

## RESPONSE

**STATUS CODE - 200:** The relay configuration associated with the given key\_stream\_id has been replaced by the data in the request

**STATUS CODE - 400:** Bad request

### RESPONSE MODEL - application/problem+json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 404:** Not found, API endpoint does not exist.

**STATUS CODE - 500:** Internal error

### RESPONSE MODEL - application/problem+json

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

## 3.3 POST /link/relay/{key\_stream\_id}

### Create link relay configuration

Create a link relay configuration record with the data in the request, using the given key\_stream\_id as an identifier

## REQUEST

### PATH PARAMETERS

| NAME           | TYPE | EXAMPLE | DESCRIPTION          |
|----------------|------|---------|----------------------|
| *key_stream_id | uuid |         | key_stream_id's UUID |

### REQUEST BODY - application/json

| NAME    | TYPE   | DESCRIPTION                   |
|---------|--------|-------------------------------|
| source* | string | URI or UUID of the source APP |

| NAME            | TYPE        | DESCRIPTION |
|-----------------|-------------|-------------|
| destination*    | array       |             |
| next_node_info* | array       |             |
| qos_request     | {recursive} | qos-data    |

## RESPONSE

**STATUS CODE - 201:** A new relay configuration has been created with the given key\_stream\_id as an identifier, from the data in the request

**STATUS CODE - 400:** Bad request

### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 500:** Internal error

### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

## 3.4 DELETE /link/relay/{key\_stream\_id}

### Delete relay configuration

Delete the relay configuration associated with the given key\_stream\_id

## REQUEST

### PATH PARAMETERS

| NAME           | TYPE | EXAMPLE | DESCRIPTION       |
|----------------|------|---------|-------------------|
| *key_stream_id | uuid |         | key stream's UUID |

## RESPONSE

**STATUS CODE - 204:** Forward table row deleted

**STATUS CODE - 400:** Bad request

### RESPONSE MODEL - application/problem+json

| NAME | TYPE | DESCRIPTION |
|------|------|-------------|
|------|------|-------------|

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 404:** Not found, API endpoint does not exist.

**STATUS CODE - 500:** Internal error

#### **RESPONSE MODEL - application/problem+json**

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

## 4. MONITOR

Generic monitoring endpoints

### 4.1 GET /monitor/capabilities

#### Get KMS capabilities

Query which essential features are supported by this instance.

#### REQUEST

No request parameters

#### RESPONSE

**STATUS CODE - 200:** The response states whether or not the KMS provides key relay support.

##### RESPONSE MODEL - application/json

| NAME                        | TYPE  | DESCRIPTION                                                                 |
|-----------------------------|-------|-----------------------------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |       |                                                                             |
| key_relay_support*          | enum  | <b>ALLOWED:</b> true, false<br>true = key relay supported, false = it isn't |
| app_api_support*            | array |                                                                             |

**STATUS CODE - 400:** Bad request

##### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 500:** Internal error

##### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

### 4.2 GET /monitor/device-info

#### Get vendor, model and version

Publish KMS instance vendor, model and version

## REQUEST

No request parameters

## RESPONSE

**STATUS CODE - 200:** The response contains the vendor, device and version

### RESPONSE MODEL - application/json

| NAME                        | TYPE   | DESCRIPTION                                                                                                                                                                                                                                                                                                |
|-----------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |        |                                                                                                                                                                                                                                                                                                            |
| vendor*                     | string | Vendor of the KMS                                                                                                                                                                                                                                                                                          |
| model*                      | string | Device brand name (model)                                                                                                                                                                                                                                                                                  |
| version*                    | string | <b>PATTERN:</b> ^(0 [1-9]\d*)\.(0 [1-9]\d*)\.(0 [1-9]\d*)(?:-((?:0 [1-9]\d* \d*[a-zA-Z-][0-9a-zA-Z-]*)?(?:\.(?:0 [1-9]\d* \d*[a-zA-Z-][0-9a-zA-Z-]*)?(?:\.(?:0 [1-9]\d* \d*[a-zA-Z-][0-9a-zA-Z-]*)?))?(\:(?:\.(?:0 [1-9]\d* \d*[a-zA-Z-][0-9a-zA-Z-]*)?)+)?)?)?\$<br>Version number in semantic versioning |

**STATUS CODE - 400:** Bad request

### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 500:** Internal error

### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

## 4.3 GET /monitor/status

### Get KMS status

Get the status from the KMS, it can be online or stopped.

## REQUEST

No request parameters



## RESPONSE

**STATUS CODE - 200:** description of the status

### RESPONSE MODEL - application/json

| NAME                        | TYPE | DESCRIPTION                                                                                                                                                                                                                                                                                         |
|-----------------------------|------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |      |                                                                                                                                                                                                                                                                                                     |
| state*                      | enum | <b>ALLOWED:</b> online, stopped<br>If the KMS is able to answer, it can only answer online or stopped. If it can't answer it is assumed to be offline, see errors below. Stopped is a state, where the KMS is alive but not operational, which could for example be a maintenance state or similar. |
| operation                   | enum | <b>ALLOWED:</b> normal operation, warnings, errors, critical<br>state if there are any operational issues. This may correlate to entries in the KMS log.                                                                                                                                            |

**STATUS CODE - 400:** Bad request

### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 404:** KMS can't be reached, therefore has to be assumed offline.

**STATUS CODE - 500:** Internal error

### RESPONSE MODEL - application/problem+json

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

## 4.4 GET /monitor/log

### Get full logging info

get the log file from the KMS

## REQUEST

No request parameters

## RESPONSE

**STATUS CODE - 200:** Log file. State the file format in the MIME type notation in the http Content-Type header. Any type can be returned, some examples are given, but it is beyond this specification to specify the log format.

### RESPONSE MODEL - \*/\*

### EXAMPLE:

"[File content of any type]"

**RESPONSE MODEL - text/plain**

**RESPONSE MODEL - application/json**

**RESPONSE MODEL - application/vnd.sqlite3**

**RESPONSE MODEL - application/octet-stream**

**STATUS CODE - 400:** Bad request

**RESPONSE MODEL - application/problem+json**

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 500:** Internal error

**RESPONSE MODEL - application/problem+json**

| NAME                        | TYPE    | DESCRIPTION                                             |
|-----------------------------|---------|---------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |         |                                                         |
| type                        | string  | Webpage with information on this problem                |
| title*                      | string  | Brief description of the problem                        |
| status*                     | integer | Response code or status code                            |
| detail*                     | string  | Information on the problem and hints on how to solve it |

## 4.5 GET /monitor/qkd-device

### Get known QKD devices and status

Receive QKD device information the KMS instance has.

### REQUEST

No request parameters

### RESPONSE

**STATUS CODE - 200:** QKD devices information correctly returned

**RESPONSE MODEL - application/json**

| NAME                        | TYPE   | DESCRIPTION                                                     |
|-----------------------------|--------|-----------------------------------------------------------------|
| OBJECT WITH BELOW STRUCTURE |        |                                                                 |
| <b>qkd_devices</b>          | array  |                                                                 |
| label*                      | string | Human readable label, for example vendor name and model number. |
| state*                      | enum   | <b>ALLOWED:</b> online, stopped, offline                        |
| interface*                  | enum   | <b>ALLOWED:</b> etsi_004, etsi_014, skip, kmip                  |
| protocol*                   | string | for ETSI GS QKD 014 and SKIP typically https.                   |

| NAME      | TYPE   | DESCRIPTION                                                                                    |
|-----------|--------|------------------------------------------------------------------------------------------------|
| peer_kms* | string | to which peer is this device connected, does not have to be UUID format, but unique in network |

**STATUS CODE - 400:** Bad request

**RESPONSE MODEL - application/problem+json**

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |

**STATUS CODE - 500:** QKD devices information could not be returned because of a server-side error

**RESPONSE MODEL - application/json**

| NAME                               | TYPE    | DESCRIPTION                                             |
|------------------------------------|---------|---------------------------------------------------------|
| <b>OBJECT WITH BELOW STRUCTURE</b> |         |                                                         |
| type                               | string  | Webpage with information on this problem                |
| title*                             | string  | Brief description of the problem                        |
| status*                            | integer | Response code or status code                            |
| detail*                            | string  | Information on the problem and hints on how to solve it |